

Homework 6

	A	B	C	D
1	Florida Power & Light peak-load pricing model			
2				
3	Input data			
4	Coefficients of demand functions			
5		Constant	Peak price	Off-peak price
6	Peak-load demand	2.253	-0.013	0.003
7	Off-peak demand	1.142	0.005	-0.015
8				
9	Cost of capacity/kwh	\$75		
10				
11	Decisions			
12		Peak-load	Off-peak	
13	Prices	\$141.78	\$80.06	
14				
15	Capacity	0.65		
16				
17	Constraints on demand			
18		Peak-load	Off-peak	
19	Demand	0.65	0.65	
20		<=	<=	
21	Capacity	0.65	0.65	
22				
23	Monetary summary			
24	Revenue	\$144.20		
25	Cost of capacity	\$48.75		
26	Profit	\$95.45		
27				

1(a) Write a mathematical programming model.

$$\begin{aligned}
 & \bullet \text{Max } X_p(2.253 - 0.013X_p + 0.003X_o) + X_o(1.142 - 0.015X_o + 0.005X_p) - 48.75 \\
 & \bullet -0.013X_p + 0.003X_o \leq -1.603 \\
 & \bullet -0.005X_p - 0.015X_o \leq -0.422 \\
 & \bullet X_p, X_o \geq 0
 \end{aligned}$$

1(b) Solve this model

ตอบ จากการแก้สมการจะได้ว่า เมื่อกำหนด capacity อยู่ที่ 0.65 เราควรตั้งราคาไฟฟ้าช่วง peak-load อยู่ที่ \$141.78 และ ช่วง off-peak อยู่ที่ \$80.06 จึงจะได้กำไรสูงสุดอยู่ที่ \$95.45

2(a) Write the problem using the matrix notation

min objective fⁿ:

$$2.253x_p - 0.013x_p^2 + 0.003x_0x_p + 1.142x_0 - 0.015x_0^2 + 0.005x_px_0 - 48.25$$

min

$$-0.013x_p^2 - 0.015x_0^2 + 0.008x_0x_p + 2.253x_p + 1.142x_0 - 48.25$$

min

$$x = \begin{bmatrix} x_p \\ x_0 \\ 1 \end{bmatrix}, Q = \frac{1}{2} \begin{bmatrix} -0.026 & 0.008 & 0 \\ 0.008 & -0.030 & 0 \\ 0 & 0 & 0 \end{bmatrix}, C = \begin{bmatrix} 2.253 \\ 1.142 \\ -48.25 \end{bmatrix}$$

quadratic term linear term

min constraint

$$-0.013x_p + 0.003x_0 \leq -1.603$$

$$0.005x_p - 0.015x_0 \leq -0.492$$

min

$$x = \begin{bmatrix} x_p \\ x_0 \\ 1 \end{bmatrix}, A = \begin{bmatrix} -0.013 & 0.003 & 0 \\ 0.005 & -0.015 & 0 \\ 0 & 0 & 0 \end{bmatrix}, b = \begin{bmatrix} -1.603 \\ -0.492 \\ 0 \end{bmatrix}$$

min matrix form: max {f(x) = 1/2 x^T Q x + C^T x : Ax ≤ b}

$$x = \begin{bmatrix} x_p \\ x_0 \\ 1 \end{bmatrix}, Q = \begin{bmatrix} -0.026 & 0.008 & 0 \\ 0.008 & -0.030 & 0 \\ 0 & 0 & 0 \end{bmatrix}, C = \begin{bmatrix} 2.253 \\ 1.142 \\ -48.25 \end{bmatrix}, A = \begin{bmatrix} -0.013 & 0.003 & 0 \\ 0.005 & -0.015 & 0 \\ 0 & 0 & 0 \end{bmatrix}, b = \begin{bmatrix} -1.603 \\ -0.492 \\ 0 \end{bmatrix}$$

2(b) Is this problem convex quadratic program (CQP)?

$$\max \{ \text{obj} = \frac{1}{2} x^T Q x + c^T x : Ax \leq b \}$$

$$\text{Hessian matrix} = \begin{bmatrix} -0.026 & 0.008 \\ 0.008 & -0.030 \end{bmatrix}$$

$$\eta_1 = -0.026 \quad (-)$$

$$\eta_2 = \det -b^2 = (-0.026)(-0.030) - (0.008)^2$$

$$\eta_2 = 0.00716 \quad (+)$$

ตอบ ปัญหานี้เป็น CQP เพราะทุก constraint เป็น linear และ Hessian matrix เป็น negative definite (เครื่องหมาย + สลับ -) เมื่อ objective function คือการทำ maximization

2(c)[Optional] Solve this problem by hand.

2(c) Solve this problem by hand

$$2.253 - 0.013x_p + 0.003x_o = 0.65 \quad (1)$$

$$1.142 + 0.005x_p - 0.015x_o = 0.65 \quad (2)$$

$$(1) \times 5; \quad 11.265 - 0.065x_p + 0.015x_o = 3.25 \quad (3)$$

$$(3) + (2); \quad 12.407 - 0.06x_p = 3.9$$

$$\text{จะได้ } x_p = 141.783 \quad (4)$$

$$\text{แทน (4) ใน (2); } 1.142 + 0.709 - 0.015x_o = 0.65$$

$$\text{จะได้ } x_o = 80.067 \quad (5)$$

$$\therefore \text{จากทุกค่าที่ได้ } x_p = 141.783 \quad \text{และ } x_o = 80.067$$

Appendix

Solver Parameters

Set Objective:

SUS14

↑

To:

☒ Max ☐ Min ☐ Value Of:

0

By Changing Variable Cells:

SB\$13:\$C\$13

↑

Subject to the Constraints:

SB\$19:\$C\$19 <= SB\$21:\$C\$21

^

↓

Add

Change

Delete

Reset All

Load/Save

☒ Make Unconstrained Variables Non-Negative

Select a Solving Method:

GRG Nonlinear

▼

Options

Solving Method

Select the GRG Nonlinear engine for Solver Problems that are smooth nonlinear. Select the LP Simplex engine for linear Solver Problems, and select the Evolutionary engine for Solver problems that are non-smooth.

Help

Solve

Close